

Interactive ELECTRICAL EQUIPMENT CONTROL Using Arduino

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This project presents a user-friendly way of controlling four electrical equipment using an LCD-based interactive menu. Usually, each equipment is controlled by a pushbutton switch. In this design, instead of four pushbuttons, only two pushbuttons are used to control the operations of four electrical equipment. Pushbuttons provide a hardware interrupt to Arduino, which executes the corresponding interrupt sub-routine to implement the control algorithm. The authors' prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the electrical equipment control system using Arduino is shown in Fig. 2. Hardware includes Arduino Uno board, 16 × 2 LCD, four-channel relay board, two pushbutton switches and a few other components.

In this project, two momentary pushbuttons (S1 and S2), namely, Next and Set, are provided so that you can select which load to control. Load number and status of each load are also displayed on the LCD menu.

On pressing Next, you can choose between various loads to control. On pressing Set, you can turn on or off a particular load.

When a particular pushbutton is pressed, its circuit generates an interrupt, which is detected by Arduino. And it executes an interrupt service routine. RC networks used in conjunction with the two pushbuttons are used for debouncing the pressed switch. Functions of the two pushbuttons are shown in the sequence of images in the table (next page).

Major components used in this project are described below.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and 14 digital I/O pins. The MCU has 32kB ISP flash, 2kB RAM and 1kB EEPROM. It can operate at a clock frequency

of 16MHz.

In this project, digital I/O pins 1, 8, 9 and 10 of Arduino are configured as output pins and are used to control the relays. Digital I/O pins 2 and 3 are configured as interrupt pins. Digital I/O pins 4, 5, 6, 7, 11 and 12 are used to interface the LCD with Arduino.

SPDT relay. A relay is an electro-mechanical switch. A four-channel relay board used in this project is shown in Fig. 3. Relays used here are of single-pole double-throw (SPDT) type. In this type of relay, output side has three terminals, namely, normally open (NO), normally closed (NC) and common (C).

When the relay is not energised,

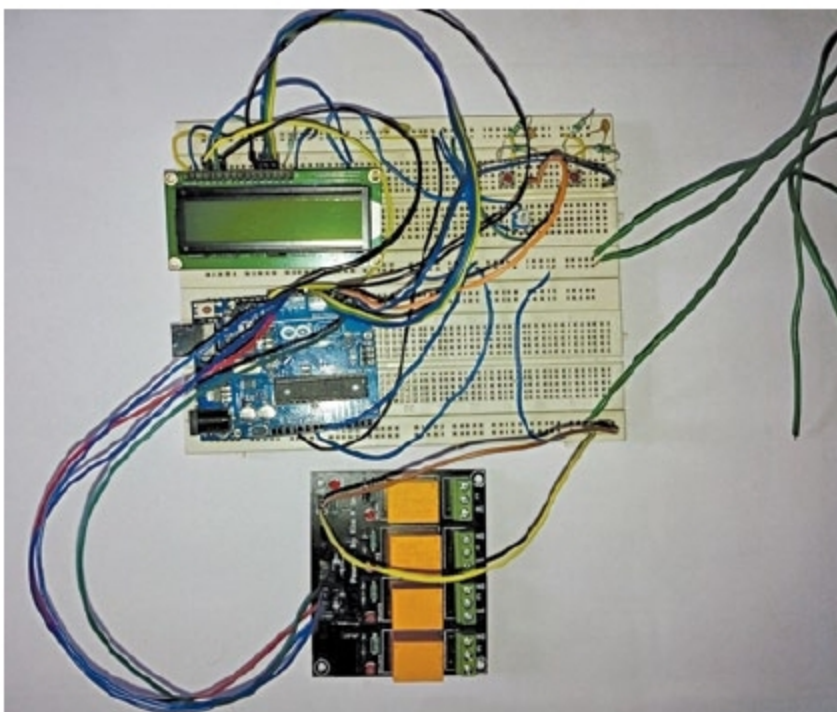


Fig. 1: Authors' prototype of electrical equipment control system

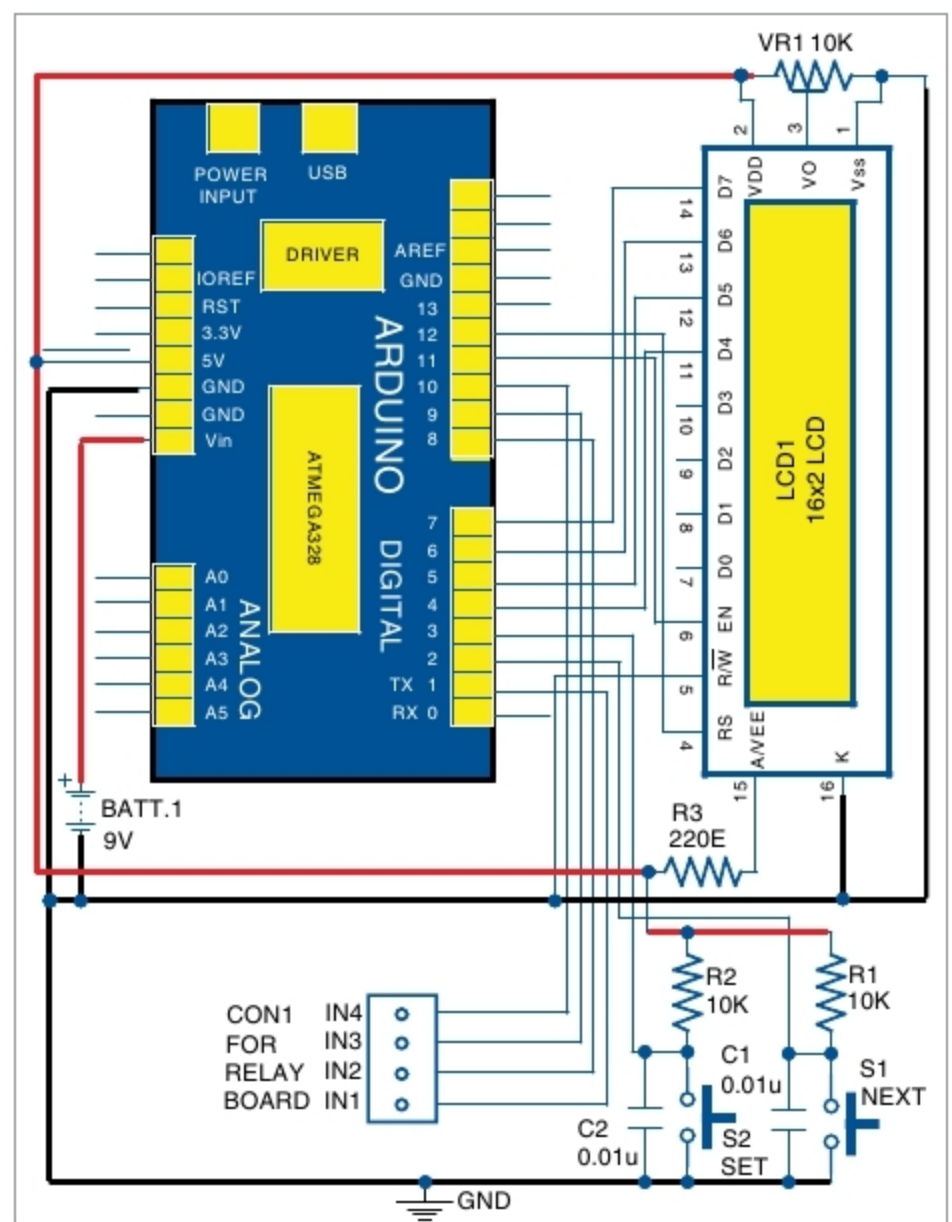


Fig. 2: Circuit diagram of electrical equipment control system using Arduino